No.

Trịnh Gia Hiệp - 2352343 - HW5 Date

1. Representation: a binary string of length $n = 12$:

$x = (x_1, \dots, x_{12}), x_i \in \{0,1\}$

- Decoding: $S = \{i : x_i = 1\}$

$k = \sum_{i=1}^{12} x_i$ (number of selected features)

The classifier is trained and validated on the columns in S only.

- Invalid state: 000000000000 ($k=0$) selects no features, so no model is trained.
- Handling: Give it the worst fitness, $f = 0$ so it never chooses a parent. Alternatively, repair it by flipping 1 random bit to 1.

2. $f(x) = acc(x) - \lambda \cdot \frac{k}{n}$, $n = 12$

- Choice: $\lambda = 0,1$: Using 12 features then costs 0,1. A feature must earn its place, but the penalty isn't large that it forces the GA to drop useful features.
- Chromosome A ($k=3$, $acc = 0,83$): $f_A = 0,83 - 0,1 \cdot \frac{3}{12} = 0,805$
- Chromosome B ($k=9$, $acc = 0,88$): $f_B = 0,88 - 0,1 \cdot \frac{9}{12} = 0,805$
- Break even analysis: $f_B - f_A = (0,88 - 0,83) - \lambda \cdot \frac{9-3}{12}$

$= 0,05 - 0,5\lambda$

$\Rightarrow$ The two chromosomes tie. $\lambda > 0,1$: A wins (I choose A because A is simpler)

$\lambda < 0,1$: B wins.

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3. Parent 1 : 101001011100

Mutation : 101001011100 → 101101011100

Crossover (one-point), parent 2 = 010110100011

	Positions 1-6	Positions 7-12
P1	101001	011100
P2	010110	100011

- Offspring 1 : P1[1-6] + P2[7-12] = 101001100011
- Offspring 2 : P2[1-6] + P1[7-12] = 010110011100

4. The symptoms (fast rise in avg fitness, collapsed diversity, stalled best fitness) indicate premature convergence.

- 2 cause :

① Selection pressure too high.

② Too little variation.

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